

Azure DevOps Basics

1. What is Azure DevOps?

Azure DevOps is a platform for managing software development using Agile/Scrum methodology. It provides tools for:

- **Work planning** (Boards, Backlogs, Sprints)
 - **Code management** (Repos)
 - **Build & Release** (Pipelines, CI/CD)
 - **Testing and Monitoring**
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2. Core Concepts We Use

Feature	What It Does
Boards	Visual interface for tracking work (Kanban style)
Work Items	Tasks, Bugs, Issues, and Epics organized into backlogs
Epics	Large features or phases of the project
Issues	Features or user stories that can be completed within a sprint
Tasks	Small chunks of work within an Issue
Sprints	1- or 2-week cycles where a set amount of work is planned and completed
Tags	Labels for filtering, grouping, and searching work items
Repos	Code storage using Git

3. How Azure DevOps Projects Are Typically Run

✅ Standard Practices:

- Backlog is filled with **detailed Issues and Tasks** across multiple sprints.
 - Work items are **estimated early**, often with story points or time-based effort.
 - Sprints are planned in advance with **strict separation of planned vs unplanned work**.
 - DevOps encourages setting **automation pipelines (CI/CD)** for testing and deployment.
 - Team velocity is tracked over time for planning future work.
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4. How Our Team Uses Azure DevOps (Notable Differences)

Area	Standard DevOps Approach	Our Team's Approach
Backlog Planning	Fill out detailed items for entire project	Mostly create Issues and Tasks for Sprint 1 + near-future
Estimation	Estimate everything upfront	Estimations added per task as we go
Epics & Sprints	Define complete Sprint Scopes early	Focus on current sprint; leave room for flexibility
Work Item Naming	Usually not prefixed	We use set conventions noted in the planning file
Review Process	Use Pull Requests & CI checks	Code review is manual for now (no CI/CD yet)
Board Columns	Often simpler (To Do / Doing / Done)	We added: To Do / Sprint Backlog / In Progress / Has Bugs / In Review / Done
Pipelines	Often automated CI/CD	We will build this later once basic model/dev is stable
Documentation/Logs	Sometimes in external tools	Ours will be stored inside DevOps or GitHub

5. Current Team Setup

- **Sprints:** We follow weekly sprints (5 total)
 - **Scrum Role Assignments:**
 - Scrum Master: [Assigned internally]
 - Product Owner: [Instructor or stakeholder role]
 - Team Members
 - **Backlog Layout:**
 - Epics define high-level phases
 - Issues = Features, deliverables, or research items
 - Tasks = Atomic work items to complete Issues
 - **Tags** used for filtering by:
 - Epic/Theme (e.g. self-study, model-dev, api)
 - Status or type (e.g. documentation, research, presentation)
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6. Things We Still Need to Add Later

- **CI/CD Pipelines**
- **Automatic testing**
- **Code review policies**
- **Sprint Burndown Charts**
- **Dashboards for velocity tracking**